Classifying Municipal Water and Sanitation Access Needs in South Africa

Introduction to Data Science (IDA117V) | Group research project | S2 2026

Statistics South Africa Census 2011 and 2022 municipal indicators

Technical report with executed analysis, saved model and local Streamlit interface

Prepared for a five-member group. Member identities and signatures must be supplied in the accompanying contribution statement.

Analysis date: 11 October 2026. The 90% access threshold is an approved project planning rule, not an official government classification.

Independent processing of Statistics South Africa data. ChatGPT assisted the implementation, analysis execution and report preparation. Group members must review the work and follow institutional disclosure rules.

1. Abstract and project overview

This project investigates municipal water and sanitation access needs using Statistics South Africa's Census 2011 and Census 2022 service-delivery tables. After removing overlapping district, provincial and national aggregates, the dataset contains 213 unique metropolitan/local municipalities across all nine provinces. A municipality has higher access need when fewer than 90% of households have improved water or improved sanitation in 2022. The rule produces 149 higher-need and 64 lower-need municipalities.

The analysis compares a majority-class baseline with Logistic Regression, Random Forest, a multilayer perceptron (MLP) and a compact residual tabular neural network. Training, validation and test partitions contain 127, 43 and 43 municipalities. Preprocessing is fitted on training data only. The residual network wins the validation comparison and receives the only hyperparameter search. Three feature-group ablations assess which information the system depends on.

The tuned residual network obtains test accuracy 0.837, macro-F1 0.811, higher-need recall 0.867, ROC-AUC 0.928 and average precision 0.970. Exact McNemar comparison with the strongest validation comparator, the MLP, gives p = 0.25. The observed improvement is not statistically significant at alpha = 0.05.

A working Streamlit interface loads the saved preprocessing-and-model pipeline, accepts municipal profiles and handles invalid inputs. The model supports a limited retrospective screening demonstration. It cannot establish present-day access, future outages or household-level need. Measured Census access/backlog indicators remain the more direct evidence for resource decisions. Spatial dependence, the small sample and the chosen label rule limit inference.

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2. Problem, stakeholder and success criteria

Municipal planners must decide where limited assessment and infrastructure resources can address service-access gaps. The affected population is households within the metropolitan and local municipalities represented in the source tables. The intended decision-maker is a municipal water/sanitation planning team supported by provincial departments. The intended benefit is a transparent screening estimate that helps identify municipalities for further assessment when target measurements are unavailable.

The prediction unit is one municipality. Inputs comprise historical service access, province, municipal category, household structure and 2022 dwelling/urban context. The classes are higher and lower 2022 access need. There is no future prediction horizon: contemporary predictors make this a retrospective classification exercise. The model must not be described as forecasting from 2011 to 2022.

Research question: Can non-target municipal indicators classify a combined water/sanitation access-need category more effectively than a majority-class baseline? Supporting questions concern traditional versus neural models, overfitting, feature-group dependence and the credibility of improvement over the strongest comparator.

The primary technical metric is macro-F1. It gives both classes equal weight despite the 149:64 imbalance. A service-delivery screening indicator is the recall of higher-need municipalities and the number missed. False negatives leave municipalities with access gaps out of an assessment queue. False positives use scarce assessment capacity on municipalities whose measured access meets the chosen threshold. Neither offline accuracy nor recall measures a realised increase in service provision.

Target rule	Definition
Higher need (1)	2022 improved water < 90% OR improved sanitation < 90%
Lower need (0)	Both percentages >= 90%
Threshold status	Operational student rule, approved before modelling
Scope	Municipal access, not water potability or continuous supply

3. Dataset provenance and responsible use

Statistics South Africa's Report 03-01-83, published 31 October 2024, compares basic-service indicators from Census 2011 and Census 2022. Its Addendum Tables provide demographic counts, dwelling counts, subsidised housing, infrastructure indices and service-access percentages. The Map Tables provide municipal categories and urban concentration, among other indicators. The report supplies definitions and contextual limitations (Statistics South Africa, 2024). Original files are preserved in the package.

The retained dataset is geographic aggregate data. It does not contain names or addresses of individual residents. The source copyright page permits processing with acknowledgement of Stats SA and a statement that results are independent processing. Sale of original or processed data requires prior permission. This is not a verified Creative Commons licence. Exact URLs and SHA-256 values are recorded in the documentation and audit outputs.

POPIA governs processing of personal information (Republic of South Africa, 2013). This project uses public municipality-level totals and requests no personal records in its interface. Future extensions using household records would require a separate lawful-access and privacy assessment. Input uploads are processed in the local application; the project does not intentionally write them to disk or publish them.

Geography and rurality may encode structural disadvantage. A higher-need label must not become a reason to penalise a community or deny services. Province/category subgroup evaluation is included, but small held-out groups cannot demonstrate fairness. Human review, measured backlog evidence and appeal/correction procedures would be necessary before an operational prioritisation system.

Boundary changes and different Census reference/collection periods restrict comparisons. Electricity access is approximated by energy used for lighting in the report. Improved water access does not prove that water is always available or safe. Improved sanitation follows the source definition, including qualifying ventilated pit latrines. The final target describes access, not reliability or disease risk.

4. Data audit and preparation decisions

Municipality code is the join key. Categories A, B1, B2, B3 and B4 identify the retained metro/local units. District and province totals overlap their constituent municipalities, so treating them as independent observations would double-count information. The cleaned table contains 213 records and 213 unique codes: 8 metros, 19 secondary cities, 26 large-town municipalities, 98 small-town municipalities and 62 rural municipalities.

All retained predictor and target cells are populated. No retained percentages fall outside 0–100, and no duplicate municipality codes occur. Source spreadsheets use merged, multirow headings and contain aggregate rows. A standard-library XLSX/XML reader converts the actual cell locations into named variables without manual editing. Checks prevent silently accepting a different row count or duplicate keys.

A source anomaly affects the 2011 dwelling categories in Addendum 2: their sum differs from Addendum 1's household count by more than 1% for every retained municipality. The report cannot justify treating those category counts as complete dwelling totals. All 2011 dwelling-category counts are therefore excluded rather than corrected with invented values. The exact reconciliation failures appear in source_anomalies.csv. The 2022 dwelling shares use their own category total as the denominator.

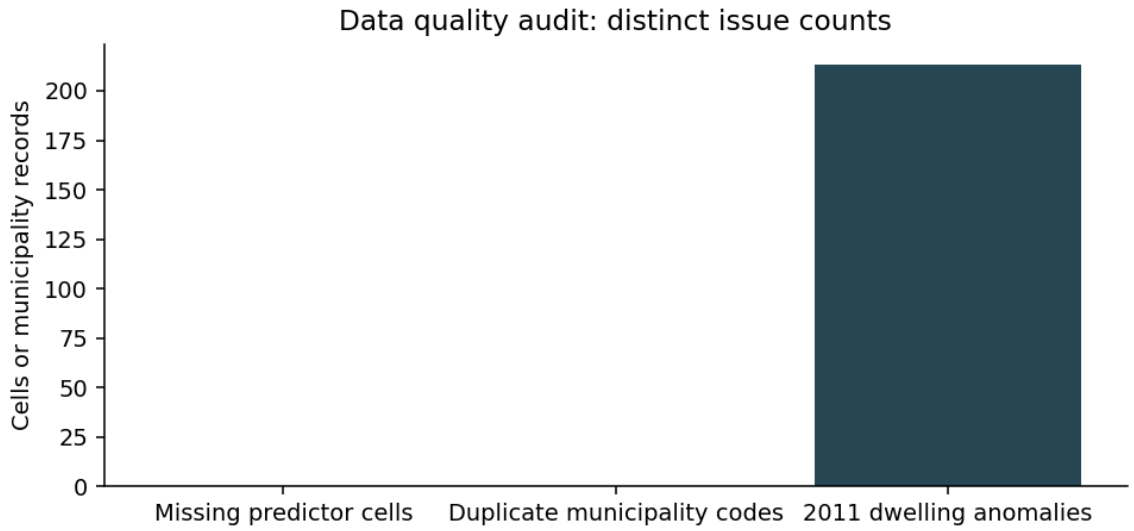


Figure 1. Distinct audit issue counts. The 213 reconciliation anomalies concern excluded 2011 dwelling data, not 213 missing model records.

Potential outliers in household counts, growth and urban concentration are retained because large cities and rural municipalities have substantively different profiles. A log(1 + households) transformation reduces count skew. No data-dependent trimming or winsorisation is performed. Data dictionary, source hashes, class distribution and range checks are included for inspection.

5. Exploratory analysis: classes and access

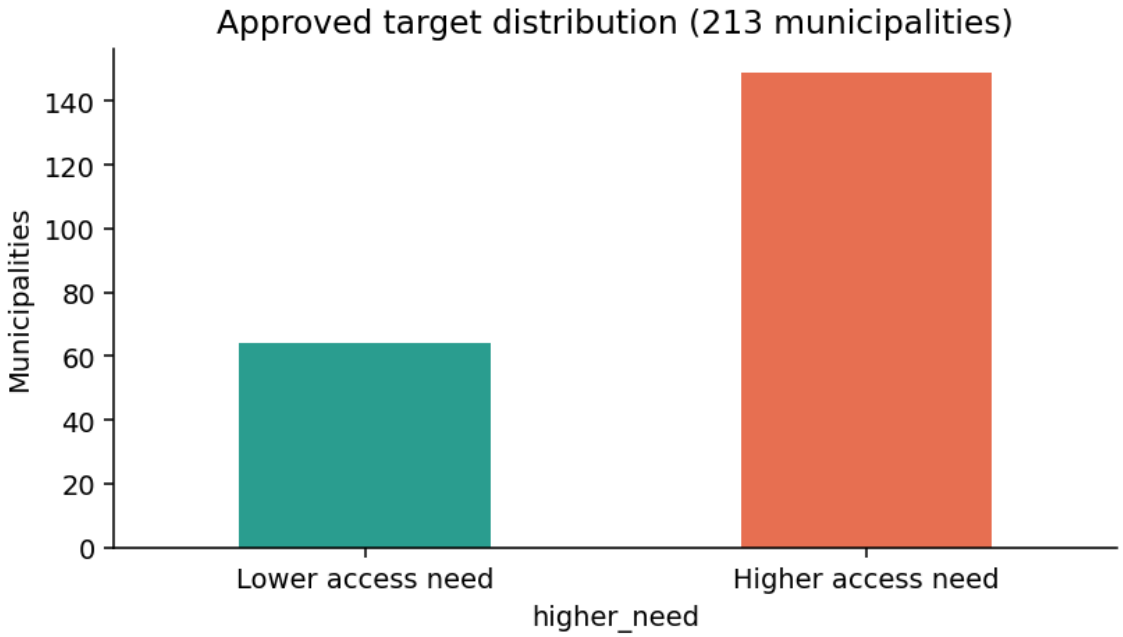


Figure 2. The approved combined target contains 149 higher-need and 64 lower-need municipalities. Counts weight each municipality equally, not each household equally.

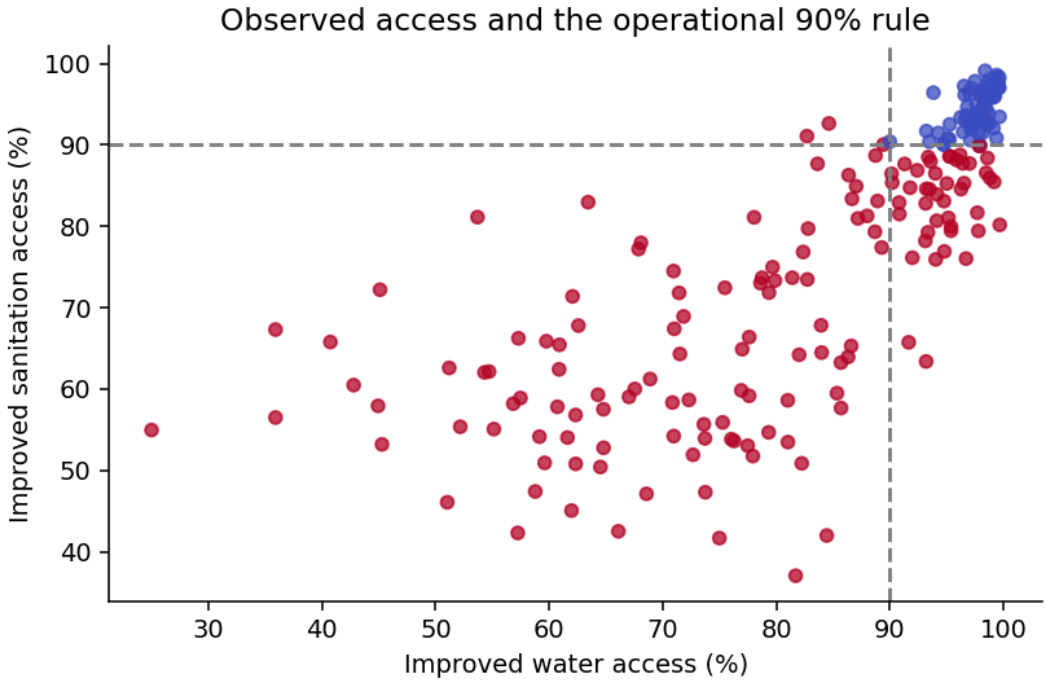


Figure 3. Each point is a municipality. The dashed 90% boundaries implement the project rule. A municipality is lower need only when both access percentages meet the threshold.

Water access ranges from 25.0% to 99.7%; sanitation access ranges from 37.0% to 99.1%. A combined label can conceal whether water, sanitation or both drive the need category, so stakeholder interpretation should always consult the observed component backlogs.

6. Exploratory analysis: relationships and geography

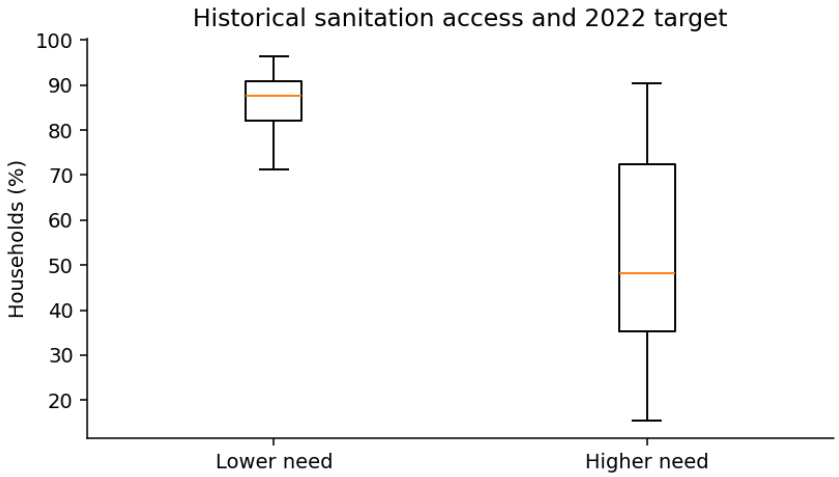


Figure 4. Historical sanitation access is generally lower in the higher-need class. This association suggests persistence in service disadvantage, without establishing a causal mechanism.

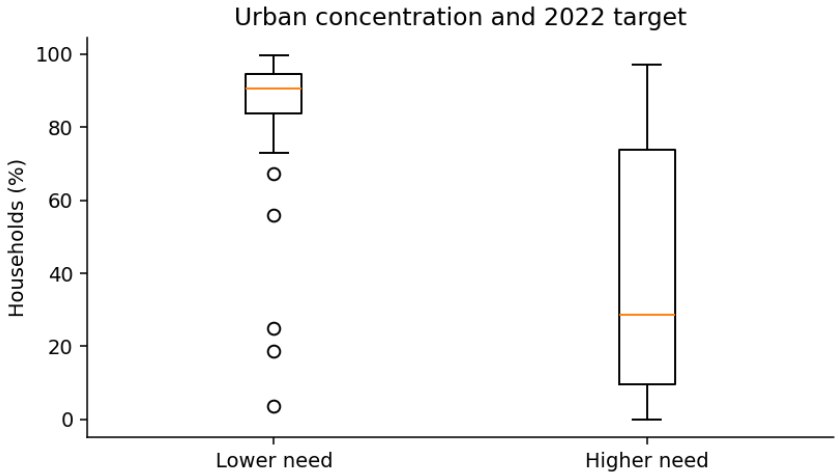


Figure 5. Urban concentration differs across classes. Geography can help prediction while also acting as a proxy for structural inequality.

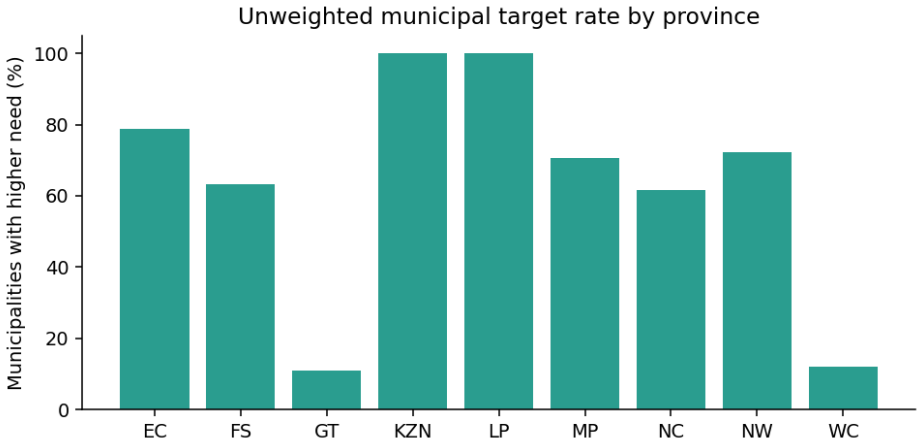


Figure 6. Unweighted municipal target rates differ by province. These are municipality proportions, not estimates of household prevalence or population service coverage.

7. Split integrity and reproducible pipeline

A fixed random seed (42) creates a stratified 60/20/20 design: 127 training, 43 validation and 43 test municipalities. Every substantive model uses the same outer partitions. The split manifest records the exact codes. The holdout is a random municipal sample, so the experiment estimates performance on similar municipalities represented by this dataset. It does not establish transfer to an entirely unseen province or later Census wave.

The sequence is: read/audit source records; construct the fixed operational label; split; fit preprocessing on training records; compare models on validation; search the selected family with training cross-validation; run validation ablations; lock models; generate test predictions once per model. Test scores never choose the family, hyperparameters, features or threshold. Re-running the supplied code repeats this fixed experiment, rather than adapting it to previous test results.

Numeric preprocessing uses median imputation followed by standard scaling. Categorical preprocessing uses most-frequent imputation and one-hot encoding with unknown categories ignored. There are no observed missing values, so imputation is a reproducible safeguard rather than a source-data replacement. Transformations that learn means, scales or categories are fitted only on the relevant training subset. This follows the leakage-prevention guidance of scikit-learn developers.

Feature engineering is deterministic: household growth is 100 × (households_2022 / households_2011 - 1), household count becomes log(1 + count), and dwelling shares use their category sum. No target-derived feature enters the predictors. Water/sanitation 2022 percentages, their backlogs, indices, composite index, names/codes and interruption fields are excluded.

Class imbalance uses balanced class weights in substantive models, computed from each training subset. Synthetic oversampling is unnecessary for this small, mixed geographic table. Four-fold stratified training CV records mean and standard deviation. Neural CV additionally splits each training fold internally for early stopping, fitting preprocessing only on the inner-fit records. Main neural fits use the fixed validation set for early stopping and model selection.

Partition	Municipalities	Purpose
Training	127	Fit and training-only CV search
Validation	43	Family selection, early stopping, acceptance and ablations
Test	43	Locked comparison and paired correctness test

8. Baseline and traditional models

The transparent baseline predicts the majority higher-need class for every municipality. It establishes what can be achieved without learning relationships between predictors and the target. On the test set it obtains 69.8% accuracy but only 0.411 macro-F1, because it never recognises the lower-need class. A high higher-need recall alone is therefore insufficient.

Logistic Regression provides a compact linear probability model with L2 regularisation and balanced class weights. The default initial C is 1, the iteration limit is 3,000 and seed 42 is fixed. It is appropriate as an interpretable traditional comparison, although correlated dwelling percentages and geographic categories complicate coefficient interpretation. The report does not claim that coefficients identify causal drivers.

Random Forest captures nonlinear relationships and interactions without imposing a sequence on the tabular features. The initial configuration uses 150 trees, maximum depth 4, minimum leaf size 5, balanced class weights and seed 42. Depth/leaf restrictions limit memorisation in the 127-record training set. Trees do not require scaling, but the shared preprocessing remains valid and keeps the compared data consistent.

Model	Validation macro-F1	Training CV mean ± SD
Dummy	0.411	0.412 ± 0.005
Logistic regression	0.795	0.878 ± 0.057
Random forest	0.823	0.870 ± 0.091
MLP	0.841	0.843 ± 0.084
Residual network	0.865	0.847 ± 0.080

The four substantive models exclude the DummyClassifier. The minimum 2+2 family requirement is met by Logistic Regression and Random Forest, together with MLP and a residual tabular network. All share the same feature set, outer partitions and 0.5 probability decision threshold.

Evidence: validation_comparison.csv, optimisation_search.json and the source implementations. CV estimates are training-only and are not interchangeable with validation or test estimates.

9. Neural architectures and training

The MLP transforms encoded features into a width-16 ReLU hidden layer, a second width-16 ReLU layer and a sigmoid output. The residual network uses a width-16 input projection and a two-layer residual branch with an identity skip connection. A final ReLU representation feeds a sigmoid output. This second architecture suits unordered tabular variables, avoiding an artificial CNN image or recurrent sequence interpretation.

Residual networks have been investigated as tabular baselines by Gorishniy et al. (2021). The supplied implementation is original educational NumPy code, inspired by the general skip-connection idea. It omits the full RTDL model's batch normalisation and other architecture details and must not be described as a reproduction of that paper.

Both neural models minimise class-weighted binary cross entropy with L2 weight regularisation. Adam uses beta1 = 0.9, beta2 = 0.999 and epsilon = 1e-8. Initial learning rate is 0.003, L2 coefficient 0.01, dropout 0.15, batch size 32 and maximum 250 epochs. Early stopping monitors unweighted validation cross entropy with patience 30 and minimum improvement 1e-5, restoring the best parameters. Seed 42 controls initialisation, dropout and batch shuffling.

The optimised residual network uses width 16, learning rate 0.001, L2 0.01 and dropout 0.1. It runs 137 epochs and restores epoch 107. There are 993 fitted scalar parameters. This capacity is substantial relative to 127 training records, so dropout, early stopping and performance-gap diagnostics are essential.

Neural gradients were checked independently using centred finite differences on a small numeric example. Maximum absolute error was below 1.5 × 10^-10 for both architectures. This supports the implementation's local derivative correctness, while empirical validation is still required to establish predictive performance.

Training uses CPU only. The recorded environment is Python 3.12.14, NumPy 2.3.5, pandas 2.2.3 and scikit-learn 1.8.0. Full package versions and fixed parameters are disclosed in requirements.txt and experiment_log.json. No GPU, pretrained model or paid service is required.

10. Locked test metrics and model comparison

Model	Accuracy	Macro-F1	Recall (1)	ROC-AUC	AP
Dummy	0.698	0.411	1.000	0.500	0.698
Logistic regression	0.791	0.766	0.800	0.890	0.954
Random forest	0.814	0.801	0.767	0.915	0.966
MLP	0.767	0.735	0.800	0.900	0.959
Residual network	0.814	0.788	0.833	0.923	0.968
Final champion	0.837	0.811	0.867	0.928	0.970

Table 1. Same 43 held-out municipalities, same target and 0.5 threshold. Recall (1) concerns higher need. AP is average precision, used as the precision-recall summary rather than trapezoidal PR area.

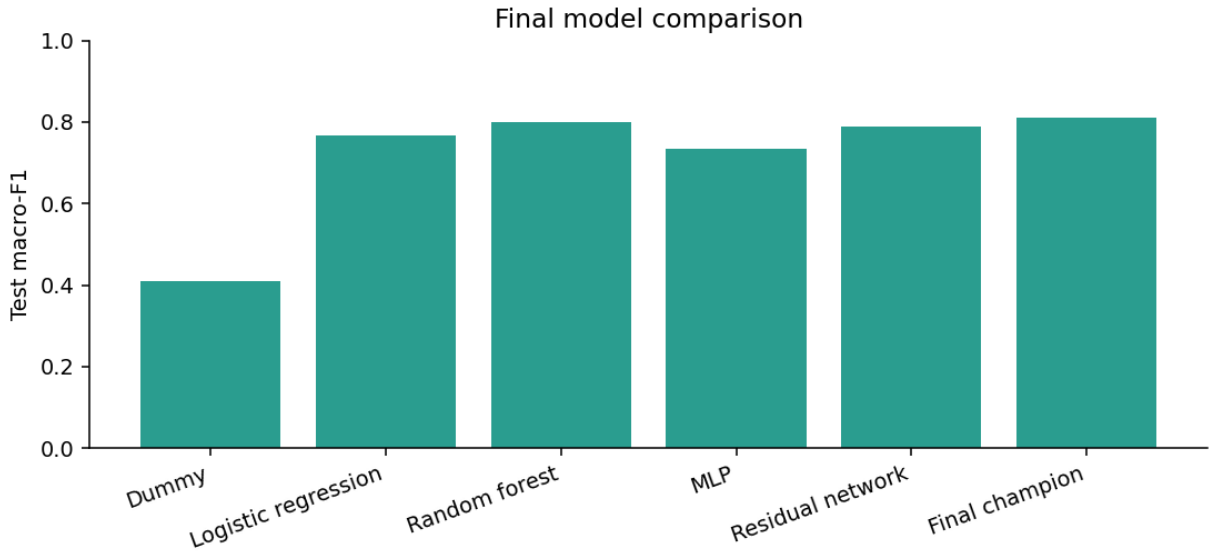


Figure 7. Final champion macro-F1 is 0.811. Random Forest obtains 0.801 on this test set, but the test result did not determine the champion or comparator.

The residual family wins the prespecified validation comparison. Its tuned configuration improves validation macro-F1 from 0.865 to 0.894. On the test set, the final model correctly classifies 36 of 43 municipalities, compared with 33 for the MLP comparator. The small difference from Random Forest demonstrates that a neural win is not universal or large.

Precision, higher-need recall, weighted-F1, ROC-AUC and average precision are included for every model in test_metrics.csv. Average precision is particularly useful with an imbalanced target. The positive class constitutes 30 of the 43 test cases. High AP does not guarantee calibrated probabilities or correct decisions at every threshold.

11. Confusion matrices, classification reports and ROC

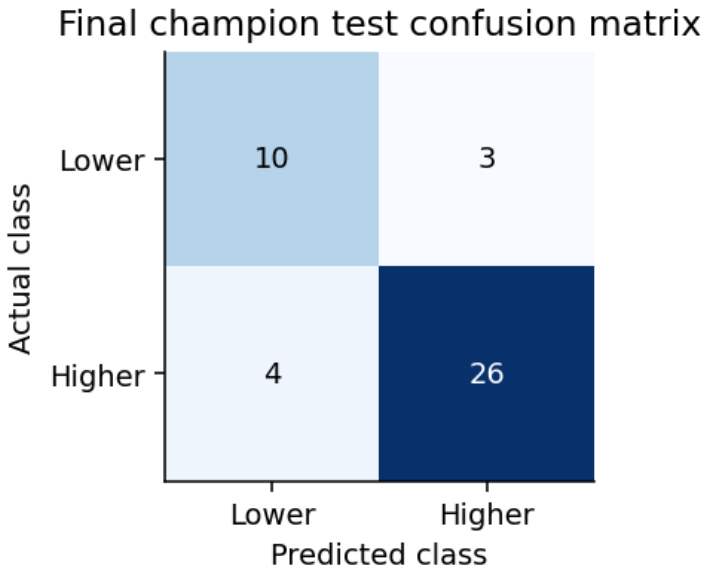


Figure 8. Final champion: 10 true lower-need cases, 26 true higher-need cases, 3 false positives and 4 false negatives.

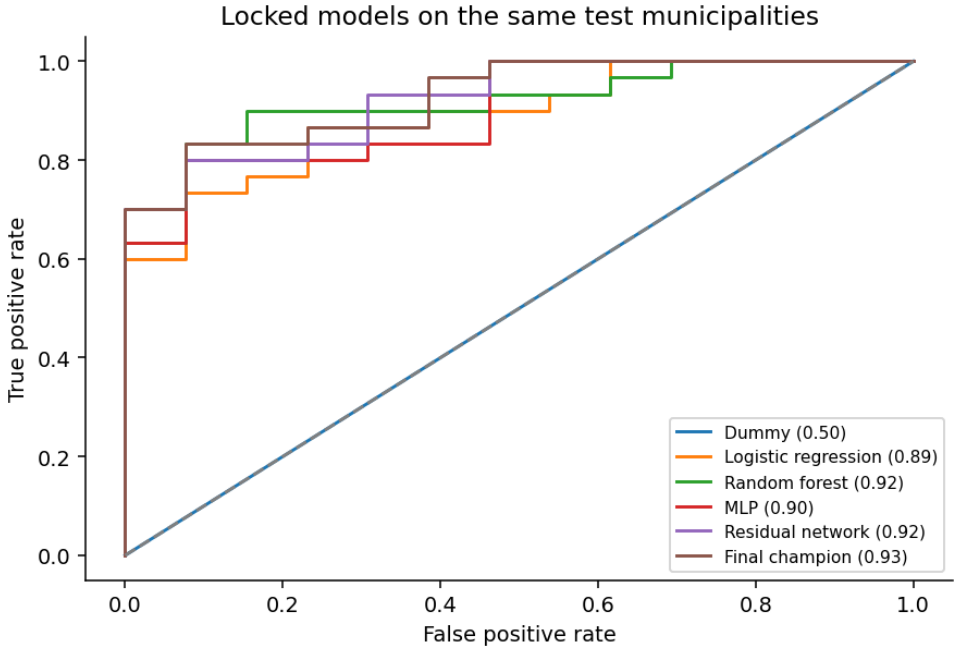


Figure 9. Common ROC curves use probability scores on identical test cases. This is binary classification, so multiclass one-vs-rest curves are unnecessary.

Four higher-need municipalities were missed. Three lower-need municipalities were incorrectly flagged. At the planning level, these errors indicate missed assessment opportunities and additional assessment costs. The data do not provide financial costs, so no monetary saving is claimed. The interface therefore presents a screening estimate and a limitation notice.

A separate labelled confusion matrix and complete per-class precision/recall/F1/support report are included for every model, including the baseline, untuned residual network and final champion. Appendix A summarises every classification report; the original JSON and CSV files preserve full precision.

12. Overfitting and uncertainty

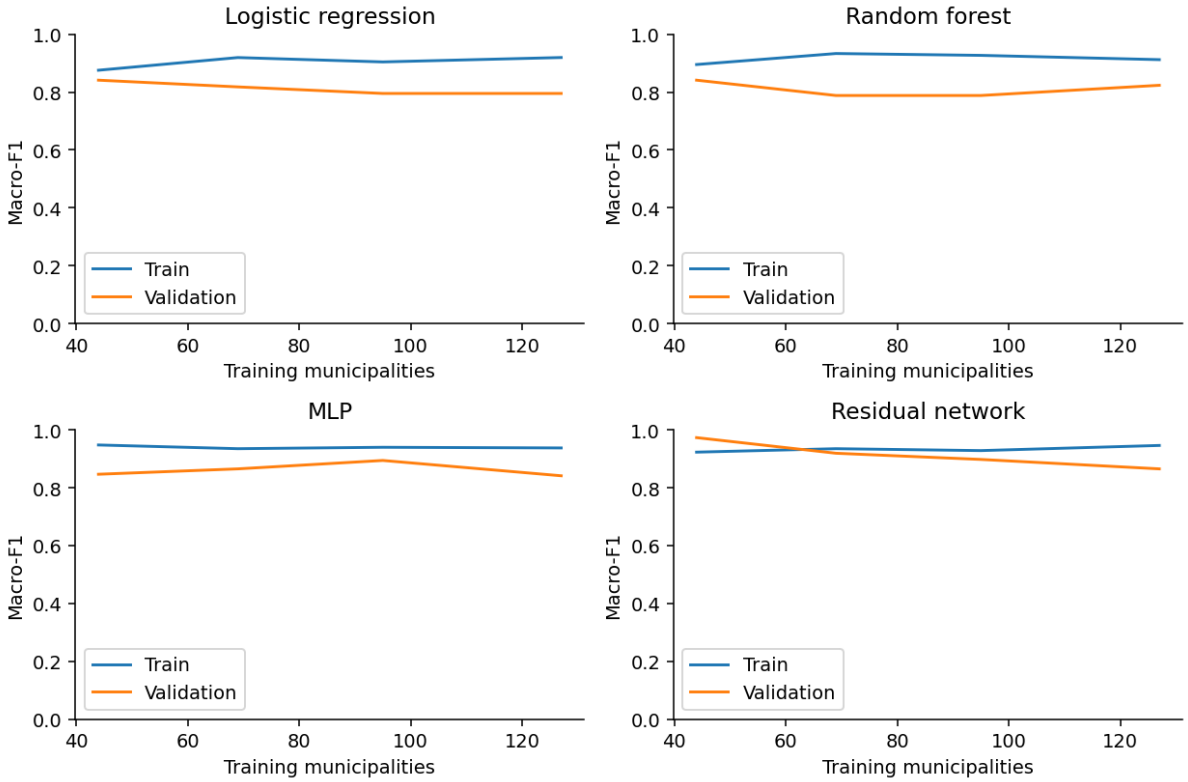


Figure 10. Training versus validation learning curves for all four substantive models. The same fixed validation set is used at four training sizes. These are diagnostic curves, not test evaluations.

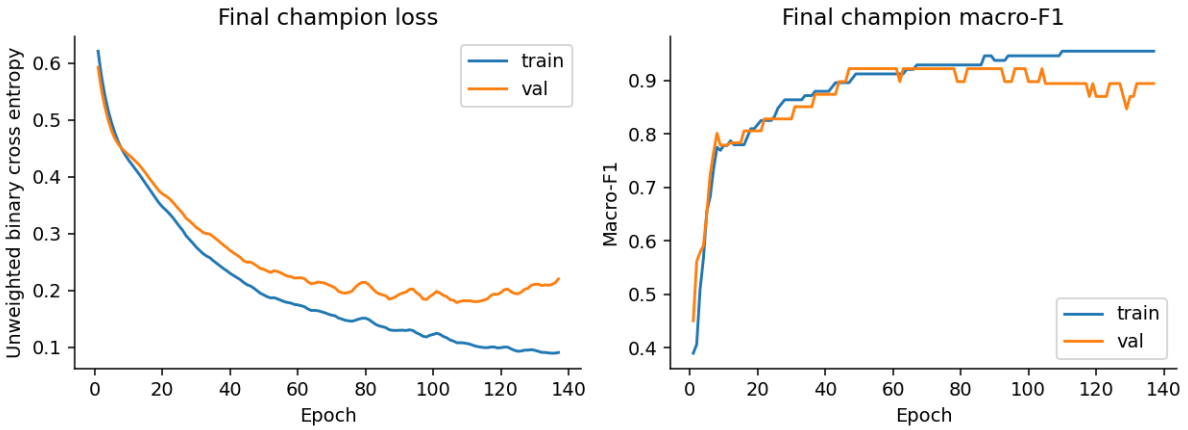


Figure 11. Final neural loss and macro-F1 curves. Loss curves are unweighted evaluation cross entropy; the optimisation objective also includes class weighting and L2.

The final network's restored training macro-F1 is 0.946, versus validation 0.894, a gap of 0.052. The initial model gaps range from 0.081 to 0.124. Regularisation reduces memorisation risk but does not eliminate generalisation uncertainty.

An exploratory 2,000-resample percentile bootstrap gives the final test macro-F1 a 95% interval of 0.669–0.926. Resamples assume independent municipalities; spatial dependence can invalidate that assumption. The interval reflects the small sample and should not be interpreted as a guaranteed operational range.

13. Champion-only optimisation

Only the residual network receives a hyperparameter search. The search includes width {8, 16}, learning rate {0.001, 0.003}, L2 {0.01, 0.1} and dropout {0.1, 0.25}: 16 configurations. Four-fold stratified training CV scores macro-F1. For neural CV, preprocessing and early stopping use only each outer fold's training subset, with a further internal early-stopping partition. The outer held-out fold remains unseen during training.

The highest mean training-CV configuration has macro-F1 0.878, SD 0.082. Chosen parameters are width 16, learning rate 0.001, L2 0.01 and dropout 0.1. Fold scores are recorded for every configuration. After fitting on the main training partition, the candidate improves fixed-validation macro-F1 and is accepted under the predeclared acceptance rule.

Stage	Validation macro-F1	Test macro-F1
Untuned residual network	0.865	0.788
Tuned residual network	0.894	0.811
Difference	+0.029	+0.023

No model is refitted on train plus validation after selection. The exact fitted pipeline accepted on validation is saved and evaluated on the test set. This sacrifices some training data but preserves the audit trail and ensures the deployed artifact matches the reported final model. No test-set threshold tuning occurs.

The improvement is small in practical terms: the tuned residual model makes one fewer error than the untuned residual network, and three fewer errors than the MLP comparator. The validation set also serves early stopping, family selection, candidate acceptance and ablations, so its performance can be optimistic. A new external or spatially separated dataset is needed before claiming operational effectiveness.

Evidence: optimisation_search.json contains the complete search grid, scores and variability; summary.json records acceptance and final metrics; final_training_details.json records best epoch, parameter count and training/validation gap.

14. Controlled ablation study

Each experiment removes one prespecified feature group from the tuned champion. Training/validation cases, seed, model settings, weighting, early-stopping rule and decision threshold remain fixed. The preprocessor is refitted only on the same training cases using the remaining features. Input dimensionality changes as a necessary consequence of removing columns. Ablations use validation data and cannot influence the final test comparisons.

Removed group	Validation macro-F1	Delta
Remove historical service indicators	0.713	-0.181
Remove geography	0.801	-0.093
Remove housing and urban context	0.805	-0.089

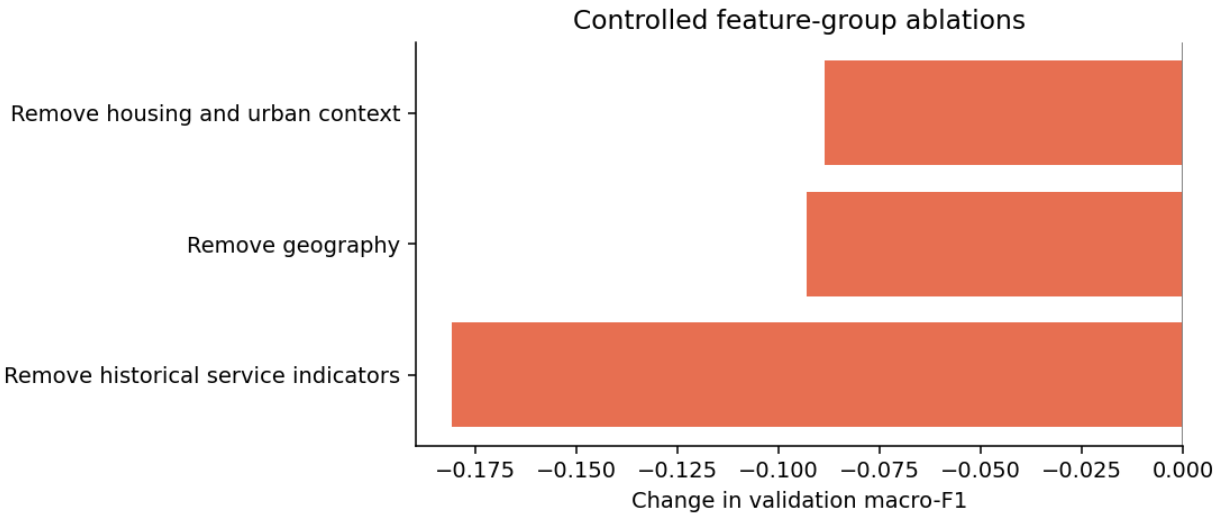


Figure 12. All three group removals reduce validation macro-F1 relative to the full champion (0.894). These effects concern information dependence under this protocol, not causal service mechanisms.

Removing the historical service indicators causes the largest decrease, approximately 0.181 macro-F1. This suggests that past service access provides important information for the constructed 2022 category. Removing geography reduces macro-F1 by 0.093, while removing housing/urban context reduces it by 0.089. Grouped removals cannot identify the unique effect of an individual correlated variable.

An ablation may also alter the early-stopping epoch or fitted representation. Holding the training procedure fixed controls the protocol, but the fitted model is expected to change. Small validation samples make the exact magnitude unstable, so these deltas should be investigated again on independent data.

15. Paired statistical validation

The strongest comparator is the MLP selected by validation macro-F1, before test evaluation. McNemar's test compares whether the final champion and comparator have equal marginal error rates on the same test cases. H0: the probability of champion-only correctness equals the probability of comparator-only correctness. H1: the probabilities differ. The significance level is alpha = 0.05.

	MLP correct	MLP wrong
Champion correct	33	3
Champion wrong	0	7

Table 2. Paired correctness table. Total 43 municipalities. Three cases are discordant: champion-only correct = 3 and comparator-only correct = 0.

With only three discordant cases, the exact binomial version is appropriate. Under H0, champion-only correct outcomes among discordant pairs follow Binomial(n = 3, p = 0.5). The two-sided exact p-value is 2 × (0.5)^3 = 0.25. scipy.stats.binomtest(3, 3, p=0.5, alternative="two-sided") reproduces this result. The continuity-corrected asymptotic statistic is 1.333, included for completeness but not used for the decision.

Because 0.25 > 0.05, H0 is not rejected. This does not prove that the models are equal; it means this small test set provides insufficient evidence of different marginal error rates. The accuracy improvement from 76.7% to 83.7% and macro-F1 difference of approximately 0.076 are observed practical differences, not statistically established gains.

McNemar concerns paired correctness, not direct testing of macro-F1 or AUC. Municipalities may be spatially dependent, which challenges the independent-pair assumption. No claim of confirmatory national inference follows from this test. The paired_test_predictions.csv file preserves every code, actual class, probability and prediction for independent verification.

16. Working interface and demonstration

The Streamlit application loads final_model.joblib, which contains the complete fitted preprocessing and model pipeline. A single-profile form accepts municipal context with guidance on categories and units. A batch mode accepts CSV using the supplied template. The application displays the predicted class, estimated higher-need probability and a short explanation of the label rule. Probabilities are explicitly uncalibrated.

The application validates required columns, province/category codes, finite numeric inputs, 0–100 percentages, dwelling-share sums and supported ranges for growth, household size and logged counts. Errors name the invalid field and explain how to correct it. The interface does not request individual personal data or expose credentials. A responsible-use notice states the historical scope and need for measured backlog evidence and human oversight.

Tested case	Predicted class	Probability
Example 1	Lower access need	3.5%
Example 2	Higher access need	96.6%
Example 3	Higher access need	95.4%

Invalid case: changing water_2011 to 150 produces “water_2011: percentages must be between 0 and 100.” Three valid form submissions and this invalid form case were exercised with Streamlit’s AppTest, and the shared inference function also tested the CSV example records. Batch-upload interaction itself requires a manual browser demonstration; its parser and inference path are included.

Local launch from the extracted project root: .\.venv\Scripts\python.exe -m streamlit run .\03_Code\app.py. The README gives complete Windows PowerShell installation steps. A hosted public demonstration is not required by the template when the launch is locally reproducible. The group must perform its live or recorded presentation demonstration.

The explanation describes the class rule and information groups, not a causal feature attribution. Profiles far outside the training distribution remain unreliable even if they pass numeric validation. Users should not upload household records or make final budget decisions from the predicted class alone.

17. Recommendations, fairness and limitations

The first recommendation is to use observed Census water and sanitation backlogs to identify access gaps. The model is a complementary retrospective screening prototype, useful when the target measurements are unavailable but historical/contextual indicators exist. A practical pilot should check current administrative data and commission field verification before allocating resources.

The target threshold materially changes class membership: at 85% access, 125 municipalities are higher need; at 90%, 149; at 95%, 188. These are label-sensitivity counts, not newly trained test results. The threshold therefore represents a planning choice with consequences, rather than a discovered scientific boundary.

Province/category metrics are available in subgroup_metrics.csv with support counts. Some test subgroups contain very few cases or no positive cases. Their measured recall is unstable or undefined, so the project cannot certify geographic fairness. Province/rurality features can amplify patterns of disadvantage. Their ablation reduces predictive performance, showing a tradeoff that should be examined before operational use.

Important limitations include only 213 observations, a 43-record test set, spatial dependence, historical Census boundaries, potentially optimistic repeated validation use, a constructed target, uncalibrated probabilities, excluded source anomalies and absence of cost/resource-capacity data. Aggregates cannot establish the circumstances of a specific household. Improved water access does not measure potability or daily reliability. Contemporary 2022 predictors prevent a genuine future forecast claim.

Do not use the system for denying services, rating individual households, current outage prediction or automatic funding allocation. Before deployment beyond a classroom demonstration, collect recent target measurements, evaluate on spatially separated municipalities or later data, review calibration and fairness, and establish a human correction process. Monitoring should track input drift, error rates by subgroup, false negatives and changes in service definitions.

18. Conclusion and reproducibility

The project demonstrates an end-to-end classification workflow using authentic Stats SA municipal data. The tuned residual network achieves test macro-F1 0.811 and accuracy 83.7%, outperforming the majority baseline and the selected MLP comparator in this split. The exact paired test gives p = 0.25, so the evidence does not establish a statistically significant advantage. Historical service information contributes strongly to the classification, as the ablation study shows.

The model's value is a limited screening demonstration, not a replacement for the source access indicators. The most defensible service-delivery recommendation is to combine measured backlogs with current evidence and human assessment. A small neural model can fit the municipal pattern, but its complexity is not itself evidence of greater public value.

The package preserves original data, a cleaned dataset, dictionary, hashes, split manifest, source code, executed notebook, model artifact, interface, requirements, figures, metric tables, classification reports, experiment/search logs and paired predictions. All source transformations are executable. Paths are relative to the project files and no secret credentials are needed.

The notebook runs the pipeline and displays results. The report and data dictionary can be rebuilt from stored outputs. The README gives Windows/VS Code execution steps. The presentation includes speaker notes and a proposed five-member speaking allocation. That allocation is a future suggestion, not an assertion of contributions already made.

Outstanding human actions: add member names/student numbers, group/lecturer details, sign the actual contribution statement, review institutional AI-assistance rules, and deliver the presentation and live/recorded demonstration. The software and analysis are complete, but signatures and demonstrated understanding cannot be supplied on behalf of students.

Mark-allocation note: Question 3's heading lists 18, although its parts and rubric total 22. Question 7's heading lists 10, although its parts/rubric total 7. The package uses the 100-mark rubric allocations: 10, 18, 22, 16, 10, 10, 7, 7. No mark or grade is guaranteed.

References

Statistics South Africa (2024). The state of basic service delivery in South Africa: Analysis of Census 2022 data. Report 03-01-83.

https://www.statssa.gov.za/publications/Report-03-01-83/Report-03-01-832022.pdf (accessed 11 October 2026).

Statistics South Africa (2024). Addendum tables, accompanying Report 03-01-83.

https://www.statssa.gov.za/publications/Report-03-01-83/ADDENDUM%20TABLES.xlsx (accessed 11 October 2026).

Statistics South Africa (2024). Map tables, accompanying Report 03-01-83.

https://www.statssa.gov.za/publications/Report-03-01-83/MAP%20TABLES.xlsx (accessed 11 October 2026).

Statistics South Africa (2024). A decade of progress in service delivery in South Africa.

https://www.statssa.gov.za/?p=17836 (accessed 11 October 2026).

Gorishniy, Y., Rubachev, I., Khrulkov, V. and Babenko, A. (2021). Revisiting Deep Learning Models for Tabular Data. NeurIPS.

https://arxiv.org/abs/2106.11959 (accessed 11 October 2026).

scikit-learn developers (n.d.). Common pitfalls and recommended practices.

https://scikit-learn.org/1.9/common_pitfalls.html (accessed 11 October 2026).

SciPy developers (n.d.). scipy.stats.binomtest documentation.

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html (accessed 11 October 2026).

Republic of South Africa (2013). Protection of Personal Information Act 4 of 2013.

https://www.gov.za/documents/protection-personal-information-act (accessed 11 October 2026).

NumPy developers (n.d.). NumPy documentation.

https://numpy.org/doc/ (accessed 11 October 2026).

pandas developers (n.d.). pandas documentation.

https://pandas.pydata.org/docs/ (accessed 11 October 2026).

Matplotlib developers (n.d.). Matplotlib documentation.

https://matplotlib.org/stable/ (accessed 11 October 2026).

Joblib developers (n.d.). Joblib documentation.

https://joblib.readthedocs.io/en/stable/ (accessed 11 October 2026).

Streamlit developers (n.d.). Streamlit documentation.

https://docs.streamlit.io/ (accessed 11 October 2026).

ReportLab developers (n.d.). ReportLab documentation.

https://docs.reportlab.com/ (accessed 11 October 2026).

Python Software Foundation (n.d.). Python 3 documentation (standard-library XLSX XML reader and notebook runner).

https://docs.python.org/3/ (accessed 11 October 2026).

Appendix A. Per-class held-out reports

Model	Class	Precision	Recall	F1	Support
Dummy	Lower need	0.000	0.000	0.000	13
Dummy	Higher need	0.698	1.000	0.822	30
Logistic regression	Lower need	0.625	0.769	0.690	13
Logistic regression	Higher need	0.889	0.800	0.842	30
Random forest	Lower need	0.632	0.923	0.750	13
Random forest	Higher need	0.958	0.767	0.852	30
MLP	Lower need	0.600	0.692	0.643	13
MLP	Higher need	0.857	0.800	0.828	30
Residual network	Lower need	0.667	0.769	0.714	13
Residual network	Higher need	0.893	0.833	0.862	30
Final champion	Lower need	0.714	0.769	0.741	13
Final champion	Higher need	0.897	0.867	0.881	30

The same 13 lower-need and 30 higher-need test municipalities appear in every report. Confusion-matrix CSVs and PNGs are in the evidence bundle. These tables support class-sensitive interpretation beyond overall accuracy.

Appendix B. Feature dictionary and compliance

Predictor	Meaning and source
province	Province code (Map/5.2 and Addendum/5)
category	Municipal subcategory (Addendum/5)
water_2011	Households with improved water, 2011 (Addendum/5 G)
sanitation_2011	Households with improved sanitation, 2011 (Addendum/5 K)
refuse_2011	Households with kerbside refuse removal, 2011 (Addendum/5 M)
electricity_2011	Households using electricity for lighting, 2011 (Addendum/5 O)
household_size_2022	Population / households in 2022 (Addendum/1 K)
formal_share_2022	Formal dwellings / sum of 2022 dwelling categories × 100 (Addendum/2 J:M)
traditional_share_2022	Traditional dwellings / sum of 2022 dwelling categories × 100 (Addendum/2 J:M)
informal_share_2022	Informal dwellings / sum of 2022 dwelling categories × 100 (Addendum/2 J:M)
urban_share_2022	Households in urban geography in 2022 (Map/5.2 Q)
rdp_share_2022	RDP/state-subsidised dwellings among eligible denominator (Addendum/3 H)
log_households_2022	Natural logarithm of 1 plus household count, 2022 (Derived from Addendum/1 J)
household_growth_pct	100 × (households 2022 / households 2011 - 1) (Derived from Addendum/1 G,J)

Percentages use 0–100 values. Urban and RDP shares retain their source meanings. The full dictionary also documents target, identifier and audit-only fields.

Appendix C. Rubric evidence map

Criterion	Marks	Evidence
1	10	Report pp. 1-2; concept_note.md; dataset_card.md
2	18	Report pp. 3-7; data_audit.json; prepare_data.py; run_analysis.py
3	22	Report pp. 8-9; neural.py; validation_comparison.csv
4	16	Report pp. 10-12; 04_Figures; classification_report JSONs
5	10	Report p. 13; optimisation_search.json; summary.json
6	10	Report pp. 14-15; ablations.csv; paired_test_predictions.csv; mcnemar.json
7	7	Report p. 16; app.py; predict.py; interface_demo.json; README.md
8	7	Report pp. 17-18; presentation; README.md; contribution_statement.pdf

Criterion 8 still requires the group’s own signed contribution statement and delivered presentation. The prepared deck, contribution form and tested local UI support those actions but do not claim they have occurred. All numerical findings above derive from the executed source code and stored output tables.